impacting world had been only a little larger, the result would have been the obliteration of the Earth. Perhaps there once were other worlds in our Solar System—perhaps even worlds on which life was stirring—hit by some demon worldlet, utterly demolished, and of which today we have not even an intimation.

The emerging picture of the early Solar System does not resemble a stately progression of events designed to form the Earth. Instead, it looks as if our planet was made, and survived, by mere lucky chance,* amid unbelievable violence. Our world does not seem to have been sculpted by a master craftsman. Here too, there is no hint of a Universe made for us.

THE DWINDLING SUPPLY of worldlets is today variously labeled: asteroids, comets, small moons. But these are arbitrary categories—real worldlets are able to breach these human-made partitions. Some asteroids (the word means "starlike," which they certainly are not) are rocky, others metallic, still others rich in organic matter. None is bigger than 1,000 kilometers across. They are found mainly in a belt between the orbits of Mars and Jupiter. Astronomers once thought the "main-belt" asteroids were the remains of a demolished world, but, as I've been describing, another idea is now more fashionable: The Solar System was once filled with asteroid-like worlds, some of which went into building the planets. Only in the asteroid belt, near Jupiter, did the gravitational tides of this most massive planet prevent the nearby debris from coalescing into a new world. The asteroids, instead of representing a world that once was, seem to be the building blocks of a world destined never to be.

Down to kilometer size, there may be several million asteroids, but, in the enormous volume of interplanetary space, even that's still far too few to cause any serious hazard to spacecraft on their way to the outer Solar System. The first main-belt asteroids, Gaspra and Ida, were photographed, in 1991 and 1993 respectively, by the *Galileo* spacecraft on its tortuous journey to Jupiter.

Main-belt asteroids mostly stay at home. To investigate them, we must go and visit them, as *Galileo* did. Comets, on the other hand,